

BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

Malaysia is home to over 1.2 million SMEs, yet the majority still manage their workforce through manual Excel spreadsheets, informal WhatsApp communications, and disconnected payroll systems. This operational blind spot costs SMEs dearly — compliance errors with statutory bodies (EPF, SOCSO, EIS, PCB) carry legal penalties, while intuition-based hiring and restructuring decisions silently drain cash flow and suppress growth.

PeopleGraph is an AI-powered Decision Support System (DSS) purpose-built for Malaysian SMEs. Rather than functioning as a static HR management tool, PeopleGraph converts fragmented operational data — payroll records, attendance logs, sales figures, and unstructured communications — into a unified **Digital Twin (Value Graph)** that directly links workforce activity to cost, productivity, and revenue outcomes.

This empowers SME owners to move beyond gut-feel management into **data-driven workforce strategy**, enabling them to simulate hiring decisions, detect hidden inefficiencies, automate statutory compliance, and receive AI-generated, actionable recommendations — all within a single platform.

The market opportunity is significant. The Malaysian SME HR technology and productivity market is estimated at **RM 35 billion**, accelerated by rising labour costs, tightening regulatory requirements, and the government's push for SME digitalisation through initiatives such as the MSME Digital Grant Madani. Within this, the addressable digital HR and analytics segment is valued at **RM 5.2 billion**, with PeopleGraph targeting a realistic **RM 45 million** capture within 18 months of launch.

PeopleGraph's competitive advantage rests on three pillars that no existing SME HR tool in Malaysia currently combines:

- **AI-Powered Decision Intelligence** — workforce insights generated by AI, moving beyond data storage into predictive, explainable recommendations
- **Automated Malaysian Statutory Compliance** — EPF, SOCSO, EIS, and PCB calculations with built-in fallback logic and strict LHDN formatting, updated modularly as regulations evolve
- **Interactive Digital Twin Simulation** — a live, what-if workforce sandbox allowing SME owners to model hiring, restructuring, and cost scenarios before committing resources

The platform monetises through a tiered SaaS subscription model, per-employee compliance processing fees, and premium AI simulation modules — projecting **RM 1.6 million in Year 1 revenue**, reaching **RM 27.5 million by Year 3** as the platform scales across major Malaysian urban SME hubs.

PeopleGraph does not just digitise HR. It gives Malaysian SME owners the same calibre of workforce decision intelligence previously accessible only to large enterprises — at a fraction of the cost and complexity.

2. Product Overview / Solution

PeopleGraph is an AI-powered HR Decision Support System that transforms fragmented SME operational data into a unified **Digital Twin (Value Graph)** — a live, interactive map linking every employee and department directly to **cost and revenue outcomes**. Rather than storing HR records passively, PeopleGraph reasons across structured and unstructured data to surface **actionable workforce intelligence** that drives smarter business decisions.

Key Features:

- **AI-powered Digital Twin Value Graph** — visualises workforce structure by linking employee activity directly to cost and revenue contribution, enabling SME owners to **simulate hiring, restructuring, and cost scenarios interactively** before committing resources
- **Automated Malaysian Statutory Compliance Engine** — calculates **EPF, SOCSO, EIS, and PCB** with strict LHDN formatting, built-in missing data fallback logic, and modular rule updates as regulations evolve
- **AI Workforce Intelligence Engine** — identifies bottlenecks, productivity sinkholes, and hidden cost drains, generating plain-language recommendations categorised as **Automate, Hire, or Outsource**
- **Multivariate Anomaly Detection System** (Isolation Forest) — flags **overtime abuse, attendance irregularities, and productivity drops** before they become costly operational problems
- **Cuti Peristiwa Shock Simulation** — calculates the financial trade-off between **statutory wage penalties vs. warehouse SLA impact** during sudden Malaysian public holidays
- **Manglish Sentiment Analysis Engine** — processes unstructured employee feedback in **Malaysia's unique code-switching communication style** to detect burnout risk and team morale trends

Technology Used:

- **Cloud-native architecture** with FastAPI backend and Next.js frontend, containerised via **Docker** for reliable and scalable deployment
- **D3.js** force-directed graph rendering powers the interactive Digital Twin with **real-time physics-based recalculation**
- **AI Agent** serves as the core AI brain for contextual reasoning, Manglish NLP, and **explainable decision generation**

- **Scikit-learn** ensemble forecasting and Isolation Forest models handle **predictive analytics and anomaly detection**
- **PostgreSQL with pgvector** enables semantic similarity search across embedded workforce and operational data

3. Market Research & Market Dynamics

3.1 Problem Landscape

Malaysian SMEs are structurally locked into **outdated, fragmented HR systems** that suppress growth and expose businesses to compliance risk. According to SME Corp Malaysia, over **98% of Malaysian businesses are SMEs**, yet the vast majority still operate without any formal HR analytics capability. Key challenges include:

- **Heavy reliance on Excel-based payroll and HR systems** — creating data silos with no real-time visibility
- **Administrative workload exceeding 30 hours per week** — spent on manual data entry, waybill processing, and payroll reconciliation
- **Frequent statutory compliance errors** across EPF, SOCSO, EIS, and PCB — carrying legal penalties under Akta Kerja 1955
- **Zero predictive capability** for hiring, workforce planning, or cost forecasting
- **Untapped unstructured data** — WhatsApp communications, informal feedback, and vendor logs contain critical operational intelligence that traditional systems cannot read

These limitations force SME owners into **reactive decision-making**, resulting in blind hiring, undetected inefficiencies, and missed revenue opportunities.

3.2 Market Niche

PeopleGraph targets a **clearly defined, underserved segment** of the Malaysian SME market:

- **Company size:** SMEs with fewer than 250 employees
- **Industries:** Retail, logistics, wholesale, and e-commerce — sectors with high operational complexity and peak-season workforce volatility
- **Decision makers:** SME owners, HR administrators, and operations managers who need decision intelligence but cannot afford enterprise-grade platforms

These users share three core priorities:

- **Compliance accuracy** — avoiding statutory penalties without hiring dedicated compliance staff
- **Cost optimisation** — identifying workforce inefficiencies before they drain cash flow
- **Affordable automation** — enterprise-level capability at SME-level pricing

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM	Total Malaysian SME Productivity & HR Technology Market	RM 35.0 Billion
SAM	SME Digital HR, Payroll & Analytics Software Market	RM 5.2 Billion
SOM	Realistic 18-month market capture (Urban SME hubs)	RM 45.0 Million

Methodology: Based on approximately 1.2 million Malaysian SMEs (DOSM 2024), with an estimated 15% in target industries, and average annual HR software spend of RM 8,000–15,000 per SME.

3.4 Market Dynamics & Placement:

- **SME Digitalisation Wave:** The Malaysian government's **MSME Digital Grant Madani** (up to RM 5,000 matching grant) and **MyDIGITAL** national blueprint are actively subsidising SME adoption of cloud-based HR and productivity tools — directly lowering PeopleGraph's customer acquisition barrier.
- **Regulatory Pressure:** The **RM 1,700 minimum wage mandate**, SOCSO wage ceiling increase to RM 6,000, and upcoming **LHDN e-Invoicing Phase 4** (July 2026, applicable to businesses under RM 1 million) are creating urgent demand for automated, accurate compliance tools that update in real time.
- **Unstructured Data Explosion:** Malaysian SMEs conduct the majority of their operations through **WhatsApp, informal chat groups, and verbal instructions** — generating vast volumes of unstructured operational data that no existing HR platform can process. PeopleGraph's **Manglish NLP engine** is purpose-built to unlock this untapped intelligence layer.

4. Value Proposition & Competitive Advantage

PeopleGraph delivers **Decision Intelligence** — not just HR management or automation. While existing tools store and organise data, PeopleGraph transforms fragmented operational data into **predictive, explainable, and actionable intelligence** that directly drives business decisions. This is the difference between knowing your payroll costs and knowing exactly which role is draining your cash flow and what to do about it.

PeopleGraph differentiates itself through **four core advantages** no existing Malaysian SME HR platform currently combines:

- **Predictive Workforce Insights** — converts payroll, attendance, and sales data into forward-looking forecasts for **hiring needs, cost efficiency, and productivity trends**, replacing gut-feel decisions with mathematically backed recommendations
- **Interactive Digital Twin Simulation** — a live "what-if" sandbox where SME owners can simulate **hiring changes, wage restructuring, or workload redistribution** and see the financial impact recalculate in real time before spending a single ringgit
- **Unified Structured + Unstructured Data Layer** — the only SME HR platform that integrates formal records (payroll, attendance, sales CSVs) with **informal operational data (WhatsApp logs, Manglish feedback)** to generate a complete, ground-truth operational view
- **Localised Malaysian Compliance Intelligence** — fully automates **EPF, SOCSO, EIS, and PCB** in strict alignment with Akta Kerja 1955, with a modular engine that updates independently as regulations change

The result: **Malaysian SME owners get enterprise-grade decision intelligence at a fraction of enterprise cost** — something spreadsheets, standalone payroll tools, and generic HR platforms simply cannot replicate.

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	High	Existing HR and payroll systems (e.g., BrioHR, Swingvy) offer compliance features but lack AI-driven decision intelligence and workforce simulation capabilities
Threat of New Entrants	Medium	High technical barrier — requires simultaneous AI integration, Malaysian compliance accuracy, and Manglish NLP domain knowledge
Bargaining Power of Buyers	High	SMEs are highly price-sensitive with low switching costs — mitigated by compliance automation as a high-value, sticky entry hook
Bargaining Power of Suppliers	Medium	Dependence on LLM APIs and cloud infrastructure — mitigated by heuristic fallback logic ensuring platform availability during outages
Threat of Substitutes	High	Excel remains free and familiar — countered by quantifiable ROI (99% payroll time saved, 30-day predictive turnover alerts)

Competitor Comparison Table:

Competitor	Strength	Weakness	PeopleGraph Advantage
Payroll Platforms (BrioHR, Swingvy)	Strong compliance automation and payroll processing	No predictive analytics , no strategic decision support	Adds AI-driven workforce intelligence + Digital Twin simulation on top of compliance
Enterprise Systems (SAP, Workday)	Highly advanced HR features and scalability	Extremely expensive , complex, not SME-friendly	Affordable and lightweight, purpose-built for Malaysian SMEs with fast onboarding
Excel / Manual HR Systems	Flexible, free, widely adopted	Error-prone, no automation, no predictive capability	Fully automated with AI analytics, compliance engine, and actionable decision insights

6. Business Model Canvas

6.1. Customer Segmentation

- **SME Owners** — require **strategic workforce insights** to reduce cost, improve productivity, and make data-driven hiring and restructuring decisions without enterprise-level tools
- **HR Administrators** — need **automated payroll processing**, statutory compliance calculations, and workforce record management to eliminate manual errors and reduce administrative workload
- **Operations Managers** — focus on **workforce efficiency, scheduling optimisation, and productivity tracking** across departments during peak and off-peak seasons

6.2. Revenue Stream

Primary Revenue — Tiered SaaS Subscription Model:

Plan	Target	Key Features
Basic	SMEs < 50 employees	Payroll automation, statutory compliance, basic dashboard
Pro	SMEs 50–150 employees	+ Anomaly detection, Digital Twin simulation, sentiment analysis
Enterprise	SMEs 150–250 employees	+ Full AI decision intelligence, custom reporting, priority support

Secondary Revenue Streams:

- **Per-employee compliance processing fees** — EPF, SOCSO, EIS, PCB automation charged per headcount processed
- **Premium AI modules** — Digital Twin simulation, workforce forecasting, and decision intelligence as standalone add-ons
- **Advanced reporting and export features** — formatted compliance exports and executive-level HR analytics for enterprise clients

6.3. Cost Structure

Cost Drivers	Cost Type
Cloud infrastructure, system development, AI model fine-tuning	Fixed Cost
LLM/API usage, compute scaling, data processing	Variable Cost
Customer support, onboarding, marketing, partnerships	Overhead Cost

6.4. Key Partners

Internal	External
Development & AI Engineering Team	SME Corp Malaysia, MRCA, FMM (adoption & outreach)
AI Agent	Government & Regulatory Bodies (LHDN, KWSP, SOCSO)
Product & Data Science Team	Cloud Infrastructure Providers (AWS / GCP)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
AI Inaccuracy in decision recommendations	High	Medium	Human-in-the-loop validation layer + rule-based fallback system
Regulatory changes in Malaysian labour laws	High	Medium	Modular compliance engine that updates rules independently
User adoption resistance (SMEs shifting from Excel)	Medium	High	Simple UI/UX design + guided onboarding + pilot trials
API or cloud service downtime	Medium	Medium	Redundant infrastructure + heuristic fallback decision logic

7. Implementation & Financials

7.1 Go-to-Market Strategy

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PeopleGraph adopts a **city-based rollout strategy**, starting with **Klang Valley SMEs** due to the highest SME density and digital adoption readiness in Malaysia. The core acquisition approach uses **compliance automation as the entry hook** — a low-friction, immediately valuable touchpoint that requires no behavioural change from the SME owner to see results.

- **3-month pilot programs** focused on payroll and statutory compliance automation — demonstrating measurable time savings before any upsell conversation
- **Partnership distribution** through **SME Corp Malaysia, MRCA, and FMM** for trust-based reach into established SME networks
- **Grant alignment** — position PeopleGraph as an eligible tool under the **MSME Digital Grant Madani 2025** (up to RM 5,000 matching), directly reducing the SME's effective cost of adoption
- **Upsell pathway** — after compliance onboarding, introduce **Digital Twin simulation, anomaly detection, and AI workforce intelligence** as the natural next step once trust is established

7.2 Implementation Plan

Phase	Action	Timeline	Target
Soft Launch	Klang Valley SMEs — Payroll & Compliance focus	Month 1-4	50 SMEs onboarded
Expansion Phase	Penang & Johor — introduce AI Anomaly Detection & Dashboard	Month 4–12	200 SMEs
Optimisation Phase	Full AI Suite rollout — Digital Twin, Simulation, NLP insights	Month 12-18	Break-even per city
TOTAL	Platform scaling, infrastructure & operations	18 Months	RM 2–3 Million investment

Use of Funds (RM 2–3 Million):

- **40%** — Engineering, AI model integration, and product development
- **25%** — Compliance database maintenance, cloud infrastructure, and scaling
- **20%** — Sales, marketing, and SME association partnerships
- **15%** — Customer success, onboarding, and operations

7.3 Revenue Projections

Metric	Year 1	Year 2	Year 3
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Active Cities	1 (Klang Valley)	3 (KL, Penang, Johor)	5 (Major Urban SME Hubs)
Active SMEs (Clients)	100	300	800
Average Revenue per SME (Annual)	RM 12,000	RM 15,000	RM 22,500
Subscription Revenue	RM 1.2M	RM 4.5M	RM 18.0M
Statutory Compliance Processing Fees	RM 0.3M	RM 1.2M	RM 4.0M
AI Decision Intelligence Add-on (Forecasting, Simulation, AI Insights)	RM 0.1M	RM 0.8M	RM 3.5M
Digital Twin / Simulation Premium Module	RM 0.0M	RM 0.5M	RM 2.0M
Total Revenue	RM 1.6M	RM 7.0M	RM 27.5M
Operating Cost	RM (2.6M)	RM (3.5M)	RM (5.0M)
Net Position	RM (1.0M)	RM (0.5M)	RM +6.5M

Revenue growth assumptions:

- Average revenue per SME increases from RM 12,000 to RM 22,500 as clients **upgrade from Basic compliance to Pro and Enterprise AI tiers**
- Operating costs grow at a **slower rate than revenue** due to cloud infrastructure economies of scale and reduced customer acquisition cost via referral networks
- Year 3 profitability driven primarily by **AI module upsell penetration** across the existing client base

7.4 Key Financial Milestones

- **Month 4:** 50 SMEs onboarded in Klang Valley — payroll and compliance live
- **Month 6:** 100 active SMEs reached — platform stability validated
- **Month 12:** Break-even achieved — expansion to Penang and Johor initiated
- **Month 24:** Cashflow positive operations — full AI suite adoption across client base
- **Month 30: Regional scaling readiness** — feasibility review for Singapore and Indonesia SME markets